

P. BHUVANESH

Robotics Engineer | ROS2 & Autonomous Systems Developer

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<https://evolve-robot-lab-1.github.io/bhuvanesh-dashboard/>

PROFESSIONAL SUMMARY

Robotics Engineer with experience in ROS2, Gazebo, RViz, URDF/XACRO, autonomous mobile robots, and embedded robotics systems. Skilled in robot simulation, TF validation, sensor integration, SLAM concepts, and autonomous navigation workflows using Linux/Ubuntu environments.

EXPERIENCE

Robotics Engineer – Evolve Robotics Lab (Mar 2025 – Present)

- Developing ROS2-based robotic software modules, Gazebo simulations, and URDF/XACRO workflows
- Working on autonomous navigation, sensor integration, TF validation, and robotics debugging

Robotics Engineer Intern – Evolve Robotics Lab (Oct 2024 – Mar 2025)

- Developed ROS2 nodes, Gazebo simulation workflows, and robot visualization pipelines
- Worked on SLAM testing, obstacle avoidance, and embedded robotics integration

TECHNICAL SKILLS

Robotics: ROS2 Humble, Gazebo, RViz, URDF/XACRO, TF2, SLAM, Autonomous Navigation

Programming: Python, C++, Embedded C, Linux/Ubuntu, GitHub

Embedded: ESP32, Arduino Uno, AVR, PIC, MSP430, Encoders, Ultrasonic Sensors, L298N

Protocols: UART, SPI, I2C

PROJECTS

ROS2 Differential Drive Autonomous Mobile Robot – Developed ROS2-based mobile robot using URDF/XACRO, Gazebo, RViz, TF validation, and encoder integration.

Autonomous Obstacle Avoidance Robot – Built autonomous robot using ultrasonic sensors, Arduino, and differential drive control systems.

Encoder-Based Mobile Robot Monitoring System – Developed encoder feedback monitoring and synchronized motor control workflows.

Robot Visualization & Simulation Workflow – Configured Gazebo and RViz environments for robotic simulation and visualization.

EDUCATION

B.E – Electrical and Electronics Engineering | Sriram Engineering College (2018 – 2022)

Diploma – Electrical and Electronics Engineering | Sriram Polytechnic College (2016 – 2018)

CERTIFICATIONS

Robotics Internship – Evolve Robotics Lab | Embedded Systems – BYTESINBITS | MSP430 Programming – Udemy | IoT Training – BSNL